|INCH-POUND|

MIL-S-13949/24A
11 April 1993
SUPERSEDING
MIL-P-13949/24
11 February 1987

MILITARY SPECIFICATION SHEET

SHEET, PRINTED WIRING BOARD, LAMINATED, BASE MATERIAL GM (WOVEN E-GLASS REINFORCEMENT, TRIAZINE AND OR BISMALEIMIDE MODIFIED EPOXY RESIN, FLAME RESISTANT, METAL CLAD OR UNCLAD)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation. MIL-S-13949

REQUIREMENTS:

Reinforcement material G in accordance with MIL-S-13949

Resin system M in accordance with MIL-S-13949

Special consideration (third letter designator) N or P in accordance with MIL-S-13949

Type of metal foil Any as specified in accordance with MIL-S-13949

Weight of metal foil. E, Q, T, H, M, 1, 2, 3, 4, 5 or 0 in accordance with MIL-S-13949.

Solderability Base materials clad with metal foil types A, B, C, D, G, H, J, K, L, and M shall be tested in accordance with MIL-S-13949 Solderability testing not applicable on all other metal foil types, unless otherwise specified in the ordering data.

Bow and twist. C or X in accordance with MIL-S-13949

Conditioning of specimens prior to test Unless otherwise specified herein, specimens shall be conditioned in accordance with the applicable test method

Property values

Dimensional stability (for base material thicknesses less than .020 inch (0.51 mm))

After bake Class A, .00075 inch/inch (0.00075 mm/mm) maximum, class B, .00050 inch/inch (0.00050 mm/mm) maximum

After stress. Class A, .00075 inch/inch (0.00075 mm/mm) maximum; class B, .00050 inch/inch (0 00050 mm/mm) maximum.

Flexural strength (for base material thicknesses greater than or equal to .020 inch (0.51 mm))

Crosswise. 50,000 pounds per square inch minimum.

Lengthwise 60,000 pounds per square inch minimum

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Flammability

Duration of burning time, each flame application

Base material thicknesses greater than or equal to .020 inch (0.51 mm)

Average. 25 seconds maximum

Single 30 seconds maximum

Base material thicknesses less than .020 inch (0.51 mm). 15 seconds maximum

Burn length (for base material thickness less than .020 inch (0.51 mm)) 12 inches (305 mm) maximum

Peel strength values for copper foil: See table I

TABLE I Peel strength values, minimum pounds per inch of width

Test and copper foil weight	Base material thickness	
	Less than 0.020 inch	0.020 inch and over
After thermal stress		
1/8-oz/ft² (designator E)	4.5	4.5
¼-oz/ft² (designator Q)	5.0	5.0
3/8-oz/ft² (designator T)	5.5	5.5
½-oz/ft² (designator H)	5.0	5.0
3/4-oz/ft² (designator M)	6.0	6.0
1-oz/ft² (designator 1)	6.5	6.5
2-oz/ft² (designator 2)	7.0	7.0
3, 4, and 5-oz/ft² (designators 3, 4, and 5)	9.0	12.0
At elevated temperature of 150°C		
1/8-oz/ft² (designator E)	3.5	3.5
¼-oz/ft² (designator Q)	4.0	4.0
3/8-oz/ft² (designator T)	5.0	5.0
½-oz/ft² (designator H)	2.0	2.0
3/4-oz/ft² (designator M)	2.5	2.5
1-oz/ft² (designator 1)	3.0	3.0
2-oz/ft² (designator 2)	3.0	3.0
3, 4, and 5-oz/ft² (designators 3, 4, and 5)	7.0	7.0
After exposure to processing solutions·		
1/8-oz/ft² (designator E)	4.5	4.5
¼-oz/ft² (designator Q)	4.5	4.5
3/8-oz/ft² (designator T)	5.0	5.0
½-oz/ft² (designator H)	4.0	4.0
3/4-oz/ft² (designator M)	5.0	5.0
1-oz/ft² (designator 1)	5.0	5.0
2-oz/ft² (designator 2)	6.0	6.0
3, 4, and 5-oz/ft² (designators 3, 4, and 5)	7.0	12.0

Arc resistance 60 seconds minimum

Dielectric breakdown (for base material thicknesses greater than or equal to .020 inch (0.51 mm))· 40 kilovolts minimum

Electric strength, average, (for thicknesses less than .020 inch (0.51 mm)). 750 volts/mil minimum

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Loss tangent at 1 MHz

Base material thicknesses less than .020 inch (0.51 mm): 0.020 maximum

Base material thicknesses greater than or equal to .020 inch (0 51 mm) 0.015 maximum

Permittivity at 1 MHz:

Base material thicknesses less than .020 inch (0.51 mm) 4.8 maximum

Base material thicknesses greater than or equal to 020 inch (0 51 mm) 5.0 maximum

Surface resistivity.

After moisture resistance conditioning 10^6 megohm minimum

After elevated temperature (125°C) conditioning 10^5 megohm minimum

Volume resistivity

After moisture resistance conditioning 10^6 megohm-cm minimum

After elevated temperature (125°C) conditioning 10^5 megohm-cm minimum.

Water absorption

Not applicable for base material thickness less than .020 inch (0 51 mm)
Base material thickness of .020 inch (0.51 mm) to .031 inch (0 79 mm) 0 80 percent.
Base material thickness of .032 inch (0.81 mm) to .062 inch (1 58 mm) 0.35 percent.
Base material thickness of 063 inch (1 60 mm) to .093 inch (2 36 mm) 0 25 percent
Base material thickness of .094 inch (2.38 mm) to .125 inch (3.18 mm) 0.20 percent
Base material thickness of .126 inch (3.20 mm) to .250 inch (6.35 mm) 0 13 percent.

Supersession information: Not applicable

Type designation: Consists of "GM" prefix followed by coded alphas and numeric characters as shown in the following example:

GM N 0620 CQ/CE B 1 C

- Base material
- Special considerations (N or P, see MIL-S-13949)
- Nominal base thickness (see MIL-S-13949)
- Type of metal foil (side 1, see MIL-S-13949)
- Weight of metal foil (side 1, see MIL-S-13949)
- Type of metal foil (side 2, see MIL-S-13949)
- Weight of metal foil (side 2, see MIL-S-13949)
- Grade of pits and dents (A, B, C, or D, see MIL-S-13949)
- Class of base material thickness tolerance (1, 2, 3, or 5; see MIL-S-13949)
- Class of bow and twist (C or X, see MIL-S-13949)

Revision letters are not used to denote changes due to the extensiveness of the changes

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CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities.
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 99
NSA - NS
DLA - ES

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5998-0024-05)

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